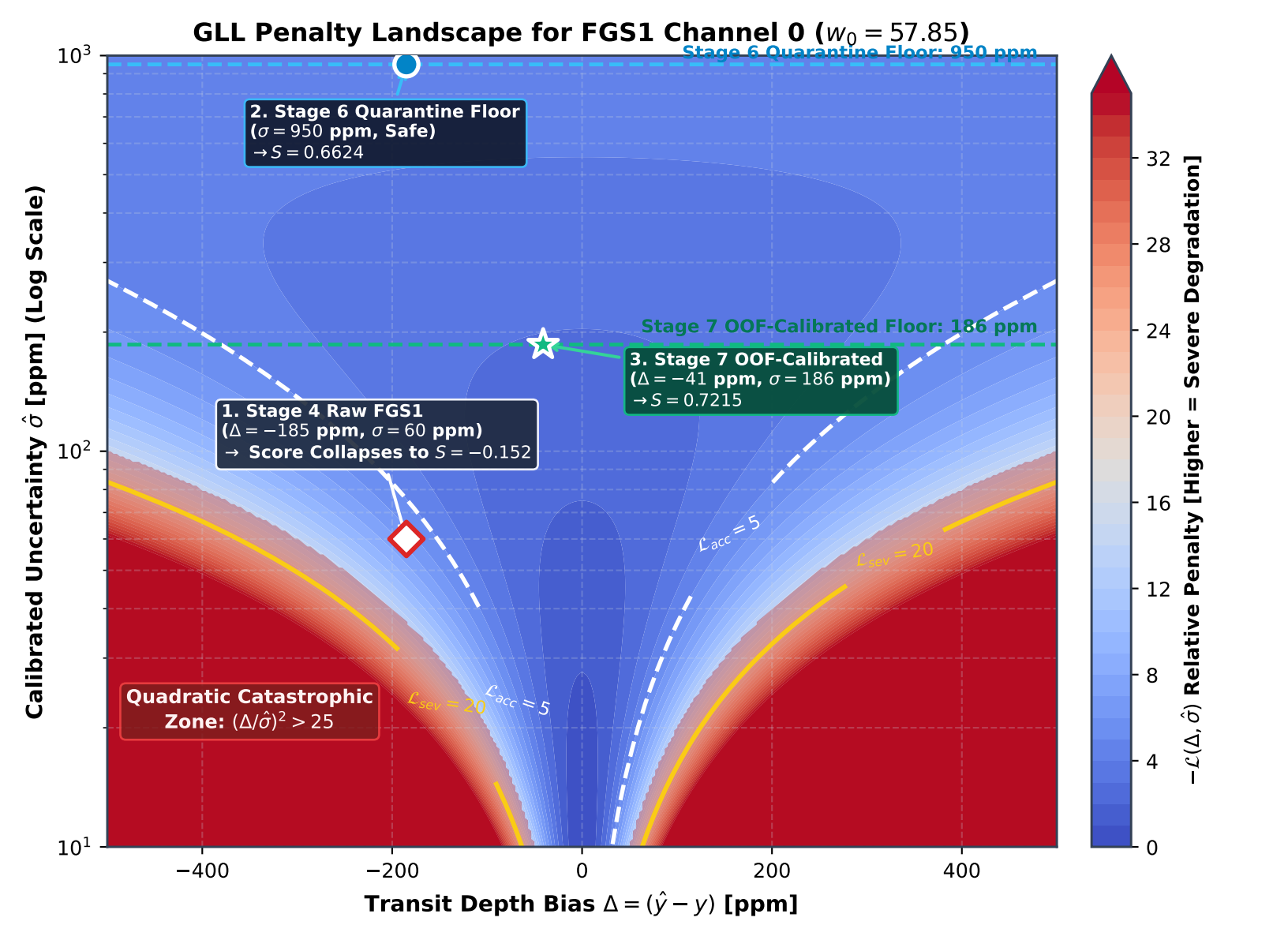

GLL Penalty Landscape for FGS1 Channel 0 ($w_0 = 57.85$)
Stage 6 Quarantine Floor: 950 ppm
2. Stage 6 Quarantine Floor
($\sigma = 950$ ppm, Safe)
$\rightarrow S = 0.6624$
Stage 7 OOF-Calibrated Floor: 186 ppm
3. Stage 7 OOF-Calibrated
($\Delta = -41$ ppm, $\sigma = 186$ ppm)
$\rightarrow S = 0.7215$
1. Stage 4 Raw FGS1
($\Delta = -185$ ppm, $\sigma = 60$ ppm)
$\rightarrow$ Score Collapses to $S = -0.152$
$\mathcal{L}_{acc} = 5$
$\mathcal{L}_{sev} = 20$
$\mathcal{L}_{acc} = 5$
$\mathcal{L}_{sev} = 20$
Quadratic Catastrophic
Zone: $(\Delta/\hat{\sigma})^2 > 25$
Calibrated Uncertainty $\hat{\sigma}$ [ppm] (Log Scale)
10^3
10^2
10^1
−400
−200
0
200
400
Transit Depth Bias $\Delta = (\hat{y} - y)$ [ppm]
$-\mathcal{L}(\Delta, \hat{\sigma})$ Relative Penalty [Higher = Severe Degradation]
0
4
8
12
16
20
24
28
32